CSE427 Section 5 Quiz 1
Fall 2025
Duration: 25 Minutes

Name: **ID:**

<u>Question</u> *[10 Points]*

Let's assume, you are given the following data and you have to apply the AdaBoost Classification Algorithm on it. X1 and X2 are the features and you have to predict the class of Y.

Observations	X1	X2	Y
1	2.5	B	Good
2	2.5	C	Good
3	3.1	B	Bad
4	3.1	A	Bad
5	3.7	A	Bad
6	4.0	B	Good

a) What will be the initial weight for all the instances? *[1 point]*
b) Using GINI Index, find out the first stump. Show your work. *[5 points]*
c) Using the stump from b), calculate the Amount of Say for the stump. *[2 points]*
d) Using the result of c), calculate the updated weights for all the instances. *[2 points]*

Solution

a) What will be the initial weight for all the instances? *[1 point]*

<u>Ans:</u> 1/6

b) Using GINI Index, find out the first stump. Show your work. *[5 points]*

<u>Ans:</u>

GINI(X1 ≤ 2.8) = 0.25
GINI(X1 ≤ 3.4) = 0.5
GINI(X1 ≤ 3.85) = 0.4
GINI(X2) = 0.222

c) Using the stump from b), calculate the Amount of Say for the stump. *[2 points]*

<u>Ans:</u> One misclassified (Obs 3). So, Amount of Say = 0.8047

d) Using the result of c), calculate the updated weights for all the instances. *[2 points]*

<u>Ans:</u>

Weight for correctly classified instances = 0.0745 (Normalized: 0.1)
Weight for incorrectly classified instance = 0.3727 (Normalized: 0.5)